

 Lex Cheat Sheet

1 Labels

Available levels: **law**, **title**, **chapter**, **section**, **article**, **paragraph**, **point**, **subpoint**

Levels required for all rules: **law**, **article**

Example:

```
law "Name of law"
article "6" "Optional title of article"
paragraph "1"
point "1"
article "7"
```

2 Types

Base types: **bool**, **int**, **float**, **string**, **span**, **time**, **money** c

Type declarations:

```
type user_consent_action
type user_id
  """a user identifier"""
type purpose is string
  """a purpose of processing"""
```

Possible values:

```
true, false           # bool
1, 18, 42              # int
1.2, 3.1415, 4.        # float
"hello", "Erika Mustermann" # string
2s, 3m, 4h, 6d, 7M, 8y # span
'2024-01-01 08:32:27'  # time
|CHF 2.|, |EUR 4000.25| # money CHF, money EUR
```

3 Events

Two keywords: **event** (for actions), **predicate** (for relations)

Capabilities: **observable**, **suppressable**, **causable**, **causable**
observable, **causable** **suppressable**, **internal**

Non-**internal** → 'brute facts' in the world
internal → 'institutional facts' defined in the law

Event declarations:

```
observable predicate PersonalData
  """data {d} is personal data of data subject {
    ds}"""
  d : data
  ds : data_subject

suppressable event ProcessesData
  """company {c} processes data {d} for purpose
  {p} as part of data processing activity {a}"""
  c : company
  d : data
  p : purpose
  a : activity

internal event IsLegal
  """processing activity {a} is legal"""
  a : activity
```

4 Expressions

(...)	Brackets
true, false, 1, 2.5, ...	Constants
=, <>, <, >, <=, >=	Relations
+, -, *, /, ^	Arithmetics
Event($t_1, \dots, t_k$)	Event/Predicate
NOT	Negation
AND, OR, IMPLIES, IFF	Connectives
EXISTS x.	There exists x...
FORALL x.	For all x...
ONCE	'some time in the past...'
HISTORICALLY	'at all times in the past...'
EVENTUALLY	'some time in the future...'
ALWAYS	'always in the future...'

5 Intervals

Temporal operators (**ONCE**, **EVENTUALLY**...) can be followed by intervals $[a, b]$, $[a, b)$, $(a, b]$, (a, b) , $[a, *)$, $(a, *)$ where $*$ = ∞ . Every a or b is a number followed by a time unit in **s**, **m**, **h**, **d**, **M**, **Y** (default: **s**).

6 Rules

6.1 Obligation rules

Meaning: 'whenever all of the premisses are true, then all the conclusions shall be true'

Syntax: **whenever**... **oblige**

```
# [GDPR Art. 5] Personal data shall be:
# 1. processed lawfully, fairly and in a
# transparent manner in relation to the
# data subject
```

```
article "5"
paragraph "1"
rule
  whenever
    ProcessesData(c, d, p, a)
  oblige
    IsLawful(a)
    IsFair(a)
    IsTransparent(a)
```

After this rule, we can add either of the two lines

```
enforceable suppressing ProcessesData
enforceable causing IsLawful, IsFair,
  IsTransparent
```

6.2 Constitutive rules

Meaning: 'whenever all the premisses are true, then all the conclusions must be the case (by definition)'

The conclusions **must be internal** events!

Syntax: **whenever**... **constitute**

```
# Processing for a purpose is lawful when
# the data subject has given consent for
# processing their data for that purpose
# (consent was given some time in the past)
```

```
rule
  whenever
    ProcessesData(c, d, p, a)
    PersonalData(d, ds)
    ONCE GivesConsent(ds, c, p)
  constitute
    IsLawful(a)
```

6.3 Exception rules

Meaning: 'whenever all the premises are true, then the rules in the conclusions shall not apply'

The conclusions **must be** labels!

Syntax: **whenever**... **except**

```
# Art. 5 par. 1 shall not apply when there
# exists a legal warrant to use the data
```

```
rule
  whenever
    LegalWarrant(c, d)
  except
    article "5" paragraph "1"
```

6.4 Patterns

Optional patterns after **whenever** and **oblige**:

once ...	'some time in the past...'
eventually ...	'some time in the future...'
... within 2d, 3M	'... within 2 days, 3 months'
... between 2d and 3d	'... between 2 and 3 days'

7 Comments

```
# This is a comment
note "This is also a comment"
```